



What are the neurological, neurovascular, and psychological responses to stretching across the general population?

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Abstract

Stretching is extensively studied and implemented to improve flexibility, as a component of warm-ups to prepare for activity, to mitigate musculotendinous injury risk, improve cardiovascular health parameters, and optimize post-exercise recovery, among others. However, less attention has focused on the effects of stretching on neurological, neurovascular, and psychological outcomes. Stress represents a pervasive element in contemporary life, with chronic mental and physical stress contributing to adverse health consequences. Evidence suggests that a bout of stretching can acutely increase sympathetic activation stimulation. However, following both acute and chronic stretching, direct and indirect measures of stress show improvements, including reductions in heart rate, blood pressure, sympathetic–parasympathetic control, cortisol release, and psychological stress (e.g., anxiety scales) along with modifications in heart rate variability and other related changes. This review synthesizes current knowledge regarding stretch-induced adaptations to neurological, neurovascular, and psychological parameters and is intended to assist researchers, health and fitness professionals, university level students, and informed fitness enthusiasts. Based on this growing literature base, we propose evidence-based practical applications that still warrant further scientific investigation.

Keywords Anxiety · Mental health · Neuromodulation · Relaxation · Static stretching · Stress · Vascular health

Introduction

Stretching has been studied for its local and non-local effects on range of motion (ROM) (Behm et al. 2021a; 2021b, 2023a), but also acute performance effects as a component of warm-up protocols (Behm et al. 2023a; Chaabene

et al. 2019; Warneke and Lohmann 2024), post-exercise recovery strategies (Afonso et al. 2021a; 2021b), impact on injury risk (Afonso et al. 2021c; Behm 2016; 2021b; 2023b; Fradkin et al. 2006; Pope et al. 1998; 2000), balance enhancement (Behm 2021b; Lohmann et al. 2024), pain modulation (Behm 2021a), and influences on muscle strength and hypertrophy (Arntz et al. 2024; Thomas et al. 2023; Warneke et al. 2023, 2024), among other outcomes. However, there are less explored avenues of research and applications that will be highlighted in this work. For example, the ability to improve ROM may depend upon increasing muscle and fascicle elongation through decreased tissue stiffness (Blazevich et al. 2014), enhancing viscoelastic stress relaxation (Magnusson et al. 1996; 1997), increasing pain and stretch thresholds (Blazevich et al. 2014; Magnusson et al. 1997), and reducing reflex activation of the musculature (Blazevich et al. 2012).

Chronic stretching improvements may be also related to enhanced relaxation, baroreflex sensitivity, and nitric oxide bioavailability (Wong and Figueroa 2021). In contrast, acute performance enhancement may involve augmented neuromuscular activation (Behm and Chaouachi 2011) and

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increased musculotendinous stiffness (e.g., decreased electromechanical delay, and briefer amortization period with stretch–shortening cycle activities) (Horita et al. 1996; Flanagan and Comyns 2008). Combined, this suggests it would be important to explore neural-mediated effects of stretching.

Stretching has also been used to induce relaxation, often coupled with breathing work (e.g., as a fundamental component of yoga or during global active stretching) (Behm 2024; Rodrigues et al. 2024; Woodyard 2011). Relaxation has been defined as “the state of being free from tension and anxiety” with synonyms listed as being calm, tranquil, or peaceful (Oxford Dictionary Online 2025). According to Turk and Winter (2005), relaxation encompasses psychological and bodily wellbeing and calmness. The caveat, of course, is that wellbeing and calmness are not perfectly defined entities, necessitating narrower concepts (e.g., arousal) to address these issues with greater scientific accuracy.

The current emphasis on techniques enhancing athletic performance has resulted in less attention to studies highlighting potential relaxation effects of stretching; effects that may extend to the broader population as well. Greater focus on the neurological, neurovascular, and psychological effects of stretching may provide insights regarding its mechanisms, while revealing potential applications that are currently overshadowed by more prominent research areas. Systemic stress experienced in daily life (Fernandez-Lopez et al. 2007) can induce neurological, neurovascular, and psychological effects such as tachycardia and arrhythmias (Buckley and Shivkumar 2016), hypertension (increased blood pressure) (Liu 2017), elevated sympathetic activation (Chu et al. 2025), and increased cortisol release (Fogelman and Canli 2018; 2019), among other responses. Relaxation techniques have demonstrated efficacy for reducing chronic pain (Vambheim et al. 2021). Also, large heart rate variability (HRV) is associated with lower heart rates, a more balanced and adaptable autonomic nervous system, and decreased cardiovascular risk and mortality. Thus, exercises such as stretching that may increase HRV may improve cardiovascular health including vascular function (e.g. decreased arterial stiffness) (Wong and Figueroa 2021; Yamato et al. 2016). One could argue that promoting interventions that reduce stress across the population (to improve overall health and reduce morbidity) should be prioritized in research agendas, even for athletic populations. Therefore, this review aims to highlight the research pertaining to the effects of stretching on neurological, neurovascular, and psychological outcomes for researchers, health and fitness professionals, university level students, and informed fitness enthusiasts. Static stretching will be a primary focus since dynamic stretching involves active movement that promotes

neuromuscular activation and overall excitation (e.g., cortical, spinal reflexive and endocrine) rather than relaxation (Behm 2024). A discussion of the effects of stretching on post-exercise recovery can be found elsewhere (Afonso et al. 2021b; 2021c).

Neurological effects of stretching

Acute neurological effects stretching

Based on stretch-induced reductions in electromyographic (EMG) activity, it is often reported that prolonged static stretching (e.g., > 60-s per muscle group) can impair muscle activation (Behm 2024; Behm and Chaouachi 2011; Behm et al. 2016; Chaabene et al. 2019; Kay and Blazevich 2012; Warneke and Lohmann 2024), suggesting reduced corticospinal excitability and reflex activity. However, as 6 sets of 45-s plantar flexor stretches did not significantly alter spinal or corticospinal excitability in the contralateral limb, global spinal motoneuron and corticospinal excitability appear unaffected (Hadjizadeh Anvar et al. 2023). Authors of one review (Trajano and Blazevich 2021) hypothesized that prolonged static stretching may influence brainstem activity to reduce monoaminergic outflow onto the spinal motoneurons. With this reduced monoaminergic drive, motoneuronal persistent inward currents (PICs) would only amplify excitatory synaptic input to a modest extent, attenuating motor output. Motoneuronal persistent inward currents (PICs) can amplify excitatory synaptic input, which could have an effect on cortically influenced and reflex influenced activity contributing to a more excited or activated neuromuscular system (Trajano and Blazevich 2021). In turn, this could inhibit attempts to relax the neuromuscular system.

Following 90 min of yoga, parasympathetic nerve activity was increased at 60 and 120 min with significant decreases in cortisol at 120 min after yoga (Eda et al. 2020); whether yoga should be considered as predominantly stretching will be addressed in Sect. [Psychological effects of stretching](#). How these acute stretch-induced effects (decreased EMG, unaffected global spinal motoneuron and corticospinal excitability, reduced monoaminergic drive, modest effects of PICs) translate to changes in physical performance remains to be determined, but so far, the impacts seem minimal (Behm et al. 2016).

Chronic neurological effects of stretching

Following 30 plantar flexor static stretching sessions, the Hoffman (H)- and Tendon (T)-reflex amplitudes were reduced, with the tonic reflex activity returning to baseline levels one month after training (Guissard and Duchateau

2004). Given the limited research, evidence for acute of chronic reflex or cortical motor neuron excitability reductions after stretching remain inconclusive.

Research has also explored how chronic stretching affects pain perception. A randomized trial with 150 sedentary office workers with neck or shoulder pain showed that a weekly 30-min stretching sessions for 4 weeks reduced perceived pain, although similar results were achieved by a Hatha yoga group (Kuptniratsaikul et al. 2023). A 4-week randomized trial with 125 women with non-specific chronic neck pain also showed that stretching had comparable efficacy to manual therapy (Ylinen et al. 2007). However, manual therapy was performed twice weekly while stretching was performed 5 times weekly, confounding data interpretation. An 8-week randomized trial showed that proprioceptive neuromuscular facilitation (PNF) stretching reduced pain when compared to a health lectures group (Gao et al. 2023), although this may simply reflect the effects of performing exercise versus no-exercise. Additionally, a 3-week randomized trial showed that stretching reduced pain in patients with hip osteoarthritis, though dry needling achieved better results (Ceballos-Laita et al. 2022). In summary, stretching may improve pain-related outcomes but is not necessarily superior to existing alternatives, and longer duration studies are warranted.

Neurovascular effects of stretching

Acute neurovascular effects of stretching

During and immediately after a stretching session, heart rate and blood pressure tend to increase (Costa et al. 2019; Siveira 2016; Yamato et al. 2016). In one study (Yamato et al. 2016), blood pressure remained elevated for 60 min after stretching (10 stretches of 30-s each) whilst heart rate declined returning to pre-stretch in that time. Feland et al. (2021) found that senior adult athletes experienced small to moderate heart rate and blood pressure increases during static and PNF stretching that recovered to baseline within 2 min of stretching cessation. Stretching large muscle groups before small muscle groups increased blood pressure more during and for 10 min after stretching than stretching small groups before large (Costa et al. 2019). Five 30-s static stretching exercises performed twice (with 30-s rests) did not alter the low-to-high frequency HRV ratio (indices of sympathetic to parasympathetic nervous activation) but improved the high frequency variability component, reactive hyperemia, and distal (foot) oxygenation, indicating better endothelial function and peripheral circulation in recovering myocardial infarction patients (mean age: 66 ± 9 years) (Hotta et al. 2013).

Perhaps more relevant than heart rate and blood pressure changes during stretching are the responses after stretching. While stretching did not increase subsequent parasympathetic activity (coefficient of variation of the R-R interval and high frequency component) in a group of 11 women, a combination of stretching and controlled breathing increased parasympathetic activity and lowered cortisol levels (Sakurai 2024). Alparslan et al. (2022) reported lower heart rates after 30-s versus 10-s of static stretching, suggesting that longer duration stretching maintains better sympathovagal balance (measured through HRV). Also, blood pressure was reduced in middle to senior aged adults following an acute bout of trunk stretching (West et al. 2024), and static stretching performed after a resistance training induced a hypotensive response from 30 to 60 min post-exercise. (Sousa et al. 2019). Two similar studies had participants perform 5 sets of 1 min triceps surae static stretches, finding that autonomic nerve activity shifted towards parasympathetic dominance and blood pressure and heart rate transiently increasing during SS and thereafter decreasing (Inami 2014; Inami et al. 2015). Furthermore, heart rate and peripheral vascular tonus significantly and immediately decreased and then returned to baseline after 5 min (5-s $\times$ 25 sets) of static stretching (Inami 2014; Inami et al. 2015), and these changes were correlated with increases in ankle ROM (Inami 2014; Inami et al. 2015). By contrast, systolic blood pressure has not been observed to change following a PNF stretching session (Cornelius and Craft-Hamm 1988; Cornelius et al. 1995).

Other studies reported post-stretching decreases in diastolic blood pressure, heart rates, and HRV ratios in young adults (Farinatti 2011), and also following 90 min of yoga (Eda et al. 2020) as well as during pregnancy (Bahadoran 2015; Logan and Yeo 2017). Compared to an acute bout of foam rolling, Theraband-assisted active static stretching induced greater post-intervention blood pressure reductions (Mevada 2024). Thirty minutes of aerobic exercise or stretching the major muscle groups similarly reduced systolic blood pressure (~ 10 mmHg) for up to 60 min post-exercise (Keller and Chilibeck 2025). A meta-analysis found acute heart rate decreases (12 studies) and a tendency for systolic blood pressure increases (13 studies) after stretching (Thomas et al. 2021). In summary, acute stretching-induced excitation or increases in neurovascular measures during stretching are accompanied by a rapid return to baseline or even bradycardia and hypotension. Stretching appears to influence neurovascular physiology, warranting increased research attention in this domain.

Chronic neurovascular effects of stretching

Hypertensive patients aged 30–70 years performing 6 months of daily static stretching, a month of rest, and

another three months of stretching did not experience significant changes in blood pressure (Yamada et al. 2022). In contrast, an 8 week stretch training program for individuals with high-normal blood pressure or stage 1 hypertension was superior to brisk walking for decreasing blood pressure (Ko et al. 2021). Three months of stretching in the morning before work (18 active and 20 passive stretches) plus progressive muscle relaxation exercises before sleeping reduced systolic and diastolic blood pressure across a 3-month post-training period (Susanto 2021). Static stretching 3 days a week for 8 weeks reduced peripheral and central blood pressure and vascular sympathetic activity in obese postmenopausal women with prehypertension and hypertension (Wong and Figueroa 2014). A similar 8 week stretch training program in obese postmenopausal women increased vagal tone and reduced sympathetic activity (Wong et al. 2017).

Male athletes with limited flexibility experienced increased HRV after 28 days of 15 min daily stretching sessions (Mueck-Weymann et al. 2004), and meta-analyses of six (20–180 stretching sessions) (Thomas et al. 2021) and eight (4–12 weeks of training) (Kato et al. 2020) stretch training studies reported significant decreases in heart rate, blood pressure, and arterial stiffness. Hence, with one exception, chronic stretch training tends to reduce heart rate, blood pressure, and sympathetic nerve activity. Longer duration, randomized studies, preferably with active controls, would allow a better assessment of the effectiveness of stretching for improving neurovascular parameters.

Psychological effects of stretching

Do the aforementioned neurolovascular and neurological effects impact psychological outcomes? Subjective measures of stress or relaxation may be considered less precise or reliable than objective measures such as HRV or blood pressure, but still modulate the more quantitative physiological measures, quality of life, and overall health. For example, greater perceived stress correlates with lower perceived quality of life (Amjadi et al. 2022) and elevated blood pressure (Miguët et al. 2023).

Acute psychological effects of stretching

Mixed acute effects of stretching have been reported. A single 15-min bout of stretching decreased state anxiety scores and increased sensation seeking scores in 14 university male and female students (Parente 2000); however, these changes were not different from the effects of aerobic exercise or a group given a lecture. The researchers also cautioned the influence of test-learning and placebo effects. In another

study, 20 male soccer players performed post-match stretching or foam rolling (Bibić et al. 2025), with foam rolling reducing perceived fatigue and stress at 2 and 24 h post-match but stretching only reducing perceived stress at 24 h post-match. Additionally, post-exercise static stretching in male and female swimmers reduced perceived fatigue *less* than static stretching combined with whole-body vibration (Merrigan et al. 2017), although it is unclear how both protocols would have compared to a passive recovery (control) protocol.

Relatedly, 90 min of reduced the overall total mood disturbance score (immediately and 60- and 120-min post-stretching) on the Profile of Mood States-Brief form as well as individual scales for tension-anxiety (60- and 120-min post-stretching), depression, and anger-hostility (120-min post-stretching) (Eda et al. 2020). Measures of affective responses for arousal and pleasure (Felt Arousal Scale and Feeling Scale) indicated that light and moderate stretch intensities during yoga resulted in perceived positive affective responses during and immediately after stretching, while more pronounced increases in affective ratings followed the higher intensity stretching (Follador et al. 2019). However, the improved feelings of pleasure and arousal after stretching, were not significantly greater than Tai Chi or yoga classes (Follador et al. 2019). Additional findings (Henriques et al. 2023) suggest that a positive affective rebound effect occurs from during, to after passive static stretching across stretch intensities and muscle groups. This positive affective rebound has also been demonstrated with aerobic activities (Dierkes et al. 2021; Ekkekakis et al. 2011), while Henriques et al. (2023) extended this phenomenon to stretching. Whereas vigorous intensities tend to result in considerable decreases in affective responses, the greater rebound with vigorous intensity stretching may be attributed to the greater discomfort of high intensity stress emanating from afferent signals (e.g., mechanoreceptors) (Henriques et al. 2023). However, this positive affective rebound should not be considered as a strategy to promote more pleasurable post-exercise feelings since the displeasure felt during exercise is associated with exercise dropout (Rhodes and Kates 2015).

However, yoga represents a multicomponent activity. While stretching plays an important role in yoga, so do muscle strength, balance, breathing, and even endurance (especially in a 90-min session). Therefore, attributing yoga effects specifically to the stretching component remains challenging. Moreover, 90-min sessions may be appropriate for the general population seeking to engage in physical activity but potentially suboptimum for athletes already engaging in large weekly training volumes. Similar confounding effects may occur with Global Active Stretching (Rodrigues et al. 2024), as this technique places considerable emphasis

on breathing work, which is known to promote relaxation (Pathan et al. 2023).

Ten minutes of whole body stretching using yoga techniques and poses without yoga style breathing and meditation techniques significantly decreased Tension–Anxiety, Depression–Dejection, Anger–Hostility, Fatigue, and Confusion scores on the Profile of Moods Scale and increased the Vigor score (Sudo and Ando 2020). There was also a cognitive enhancement with a decrease in Stroop interference performance. A significant negative correlation between Δ Vigor score and Δ Stroop interference suggested that a Vigor score increase was associated with higher cognitive performance. Thus, acute stretching improved mood states and cognitive performance in physically inactive people (Sudo and Ando 2020). Another study by Sudo et al. (2015) using 30 min of static stretching found a decreased reaction time in a visual search task which was accompanied by an increase in positive engagement and a decrease in anxiety. Whilst there was no modulation in cerebral hemodynamics or oxygenation, they proposed that improved response speed was associated with an increase in positive engagement and a decrease in anxiety after static stretching. These acute psychological responses to stretching might be attributed to stretching-induced effects on brain activity.

Both high and low intensity stretching (5 sets of 120 s each with 30-s rest periods) induced lower alpha and beta brain wave activity and increased parasympathetic activity while static stretching (Imagawa et al. 2023). On the contrary, a combination of resistance training and stretching enhanced sit and reach range of motion and also improved athletes' psychophysical self-awareness (exercise-related bodily sensations) more effectively than static stretching alone (Esposito et al. 2025). Therefore, stretching may potentially exert acute positive effects on psychological outcomes that are not necessarily superior to other interventions. Research on this topic remains limited, particularly in athletic populations.

Chronic psychological effects of stretching

Corey et al. (2014) conducted a 1-year multi-center randomized controlled trial with 171 healthy participants comparing restorative yoga with static stretching. The stretching group demonstrated decreased cortisol at waking and bedtime, greater reductions in chronic perceived stress severity, and perseverative thoughts about their stress and increases in social support compared to the restorative yoga group. Eight weeks of high intensity interval training or stretching increased perceived joy (Trier Social Stress Test) while decreasing tension and elevating perceived levels of general self-efficacy (Javelle et al. 2024). Twelve weeks of 10-min stretching sessions after work reduced perceived

anxiety, pain, and exhaustion compared to workers receiving no intervention (Montero-Marín et al. 2013). However, because the controls were passive, it remains unclear whether improvements were related to stretching or to physical exertion generally.

Eight weeks of neuromuscular training (balance, strength, plyometric, and change-of-direction exercises) reduced anxiety and improved self-confidence in highly trained pubertal male soccer players whereas static (4 weeks) followed by dynamic (4 weeks) stretch training improved mental wellbeing (Hammami et al. 2025). Perceived anxiety in office workers decreased after 4 weeks of stretching, comparable to effects in a hatha-yoga group (Kuptniratsaikul et al. 2023). Therefore, the chronic psychological effects of stretching remain underexplored in the literature.

Research opportunities for potential applications

The neurological, neurovascular, and psychological effects of stretching likely depend on a complex interplay between training history, previous knowledge, expectations, and information presentation; collectively, the placebo effect may modulate intervention effectiveness (Lebeau et al. 2024). Given the scarcity of robust evidence for stretch-modulated relaxation (both acute and chronic), especially in athletes, the opportunities mentioned here are speculative and require research confirmation.

Early morning static stretching routine: A stretching routine upon waking may help start the day feeling relaxed. We do not advocate that stretching will universally benefit everyone or demonstrate superiority to other interventions (e.g., yoga). However, stretching represents a practical option requiring no special equipment and offering “flexibility” (no pun intended) in adjusting duration and intensity based on an individual's state upon waking and time available.

Post-exercise stretching: While post-exercise stretching does not appear to enhance recovery (Afonso et al. 2021b), it may improve perceived relaxation (e.g., reduced stress). Without clear evidence demonstrating stretching's superiority to other interventions, the approach to stretching could depend on individual preference.

Pre-sleep stretching routine: Pre-sleep stretching could potentially improve sleep duration and quality, but researchers must control numerous potential confounders: people may perform pre-sleep stretching accompanied by reduced and/or relaxing noise or music, dim lighting, and other co-interventions. Without appropriately designed studies, misinterpretations may proliferate. Pre-sleep stretching may also reduce the frequency and severity of nocturnal leg cramps in people who have them regularly (Hallegraeff et

al. 2012). Longer-term, low-intensity stretching has been shown to improve sleep quality in sedentary, overweight, postmenopausal women (Tworoger et al. 2003), but these results apply to a specific sample and do not specifically address pre-sleep stretching. A review showed little evidence for chronic stretching effects on sleep quality in people with sleep disorders (Mohammad et al. 2024). Overall, the acute effects of pre-sleep stretching on sleep quality are underexplored.

Pre-match stretching: Even without consequential physical performance effects, participants may express greater confidence in performance when stretching is included in the warm-up (Blazeovich et al. 2018). This potentially reflects a placebo effect (i.e., due to traditional views on pre-exercise stretching) that might prove beneficial, particularly before official competitions. Some athletes experience excessive anxiety before competitions and may therefore benefit from pre-competition relaxation techniques, including static stretching as a viable option. Although prescribing static stretching universally before a competition may be (potentially) inappropriate, individual athletes may benefit from it. Whether this should occur immediately pre-competition or a few hours before remains an open question.

Discussion

Stretching is widely implemented, but research frequently focuses on performance effects and underlying biomechanical and physiologic adaptations (Afonso et al. 2024; Wohmann et al. 2024). Considerably less is known regarding the neurological, neurovascular, and psychological effects of stretching. This evidence gap provides exciting opportunities for future research. Rather than continuing to devote extensive attention to the effects of stretching on ROM, strength, and other performance parameters, researchers could expand their exploration of neurological, neurovascular, and psychological effects. Most studies in these domains are *not* performed with athletes, indicating a clear research opportunity. Figure 1 synthesizes key aspects of this perspective article.

Since there are only a limited number of studies in these areas, and some provide only indirect evidence (e.g., studies implementing yoga, which involves more than just stretching), highlighting possible dose–response patterns would be difficult and perhaps premature. In fact, dose–response relationships for stretching are only available for a very select sub-set of applications, and even then, only for very specific populations.

Whereas static stretching is generally considered safe when appropriate durations and intensities are implemented

Fig. 1 Neurological, neurovascular, and psychological effects of static stretching and opportunities for potential applications

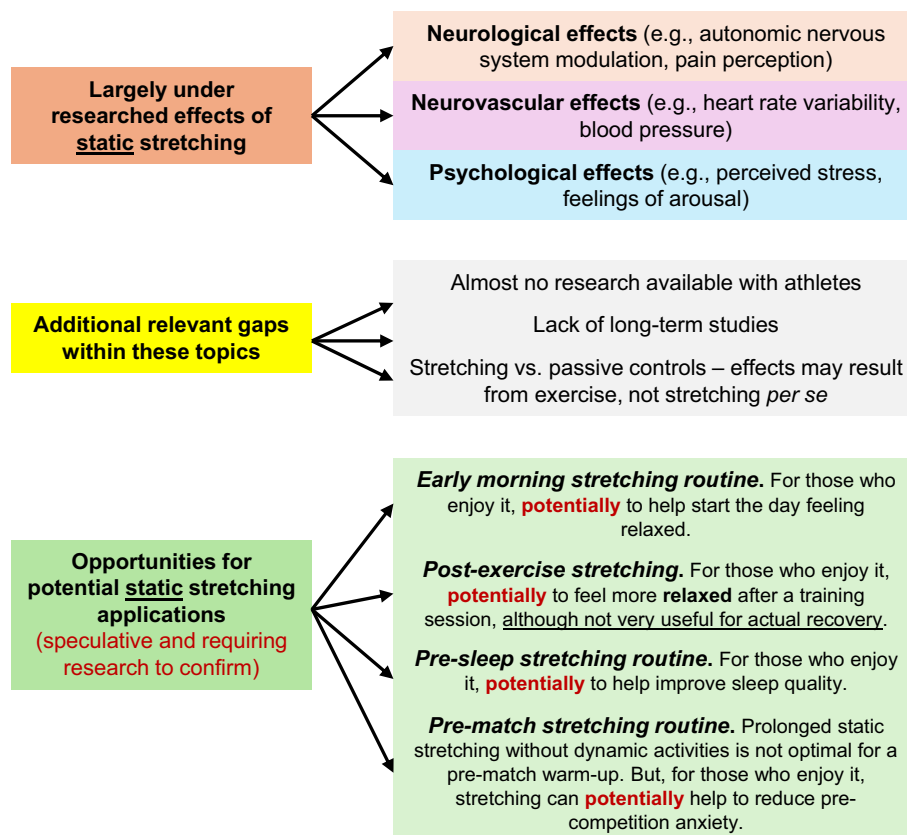


Table 1 Acute and chronic responses to stretching for neurological, neurovascular and psychological responses

Table 1 Acute and chronic responses to stretching for neurological, neurovascular and psychological responses	Acute responses		Sample references	Chronic responses	Sample references
	Neurological	↓EMG	Behm and Chaouachi 2011; Behm et al. 2016; Chaa-bene et al. 2019; Kay and Blazevich 2012; Warneke and Lohmann 2024	↓H- and T- reflexes	Guissard and Duchateau 2004
		↓Monoaminergic drive	Trajano and Blazevich 2021	↔ Cortical and reflex excitability	Behm 2024
		↔ Spinal motoneuron and corticospinal excitability	Hadjizadeh Anvar et al. 2023	↓Pain-related outcomes	Kuptniratsaikul et al. 2023, Ylinen et al. 2007, Gao et al. 2023, Ceballos-Laita et al. 2022
	Neurovascular	↓PICs	Trajano and Blazevich 2021		
		↑HR and BP during and immediately after stretching	Costa et al. 2019; Siveira 2016; Yamato et al. 2016; Feland et al. 2021	↓HR, BP, sympathetic drive, arterial stiffness	Ko et al. 2021; Susanto 2021; Wong and Figueroa 2014; Wong et al. 2017; Thomas et al. 2021; Kato et al. 2020
		After stretching, ↓HR, BP, cortisol, vascular tonus ↑Parasympathetic	Sakurai 2024; Alparslan et al. 2022; West et al. 2024; Sousa et al. 2019; Inami 2014; Inami et al. 2015; Farinatti 2011; Mevada 2024; Keller and Chilibeck 2025; Eda et al. 2020		
	Psychological	↓Perceived fatigue and stress	Bibić et al. 2025; Merrigan et al. 2017	↓Cortisol,	Corey et al. 2014
		↓Total mood states score (POMS)	Eda et al. 2020; Parente 2000; Sudo and Ando 2020	↓Perceived stress, anxiety and pain	Javelle et al. 2024; Montero-Marín et al. 2013; Kuptniratsaikul et al. 2023
		↑Affective responses (i.e., Felt Arousal and Feeling scales)	Follador et al. 2019, Henriques et al. 2023	↓Exhaustion	Montero-Marín et al. 2013
↑Parasympathetic drive		Imagawa et al. 2023			
	↑Cognitive performance	Sudo and Ando 2020, Sudo et al. 2015			

(Behm 2024; Behm et al. 2016, 2021a, 2021b), the use of stretching should be cautioned with individuals experiencing acute muscle/tendon injuries, hypermobility or unstable joints (especially with a history of dislocation), inflammatory joint diseases (especially during flare-ups), nerve irritation, or with certain positions (e.g., hip flexion) in pregnant women during the last trimester. Moreover, rare adverse effects have been described even in sports practitioners, such as a single published case of sciatic neuropathy (Shim et al. 2013).

Care must be taken with regard to study design: (i) studies comparing stretching interventions to passive controls allow conclusions about exercise versus no-exercise but perhaps not about stretching specifically; (ii) comparing

stretching to alternative interventions (e.g., strength training) would provide better insights into its effectiveness; and (iii) multimodal interventions (e.g., yoga) are likely suboptimal for isolating the effects of stretching. Current research remains scarce and heterogeneous; the few randomized studies available show that stretching *may* produce positive neurological, neurovascular, and psychological outcomes, but not necessarily superior to alternative interventions. A summary of the potential or possible outcomes is available in Table 1.

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Data availability Not applicable—review paper.

Declarations

Conflict of interest The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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